the pH range 2–6. From pH6 to 10.2 the mobilities of the protein-coated high-density particles accelerated to a level 35% greater, at pH8.2, than the low-density particles.

It is difficult at this stage to provide a complete explanation of this electrophoretic behaviour, but we consider it likely that on the smooth flat faces of the high-density particles, the protein is adsorbed as an expanded monolayer and allows a higher particle mobility from pH6 to 10.2, owing to the exposure of charged side groups (Chattoraj & Bull, 1959). In contrast, the uptake of adsorbed protein within the porous aggregates of low-density $BaSO_4$ (Howell & Deacon, 1975) prevents the increased mobility owing to unfolding of the protein. If this interpretation is valid, it would support the proposal of Howell & Dupe (1972) that partial unfolding on this adsorbent encourages autoproteolysis of factor X leading to its activation.

For protein adsorbed to particles of barium citrate the pH–mobility curve showed a slight displacement in the acid region compared with the bare particles and reflects the small amount of protein adsorbed.

The pH–mobility curve obtained for barium citrate precipitated *in situ* was interesting in that the linear variation of mobility with pH is typical of non-ionogenic particles (Goddard, 1970). This supports the proposal of Tishkoff *et al.* (1968) that the protein is sequestered within a crystal lattice, and in this state, it is unlikely that any unfolding and activation can occur.

S. L. M. D. acknowledges receipt of a Science Research Council studentship.

Chattoraj, D. K. & Bull, H. B. (1959) *J. Am. Chem. Soc.* **81**, 5128–5136
Goddard, G. H. (1970) Ph.D. Thesis, University of London
Howell, R. M. & Deacon, S. L. M. (1975) *Thromb. Diath. Haemorrh.* **33**, 256–270
Howell, R. M. & Dupe, R. J. (1972) *Thromb. Diath. Haemorrh.* **28**, 306–316
Howell, R. M. & Scott, G. B. D. (1964) *Br. J. Exp. Pathol.* **45**, 618–626
Lowry, O. H., Rosebrough, N. J., Farr, A. L. & Randall, R. J. (1951) *J. Biol. Chem.* **193**, 265–275
Stenflo, J., Fernlund, P., Egan, W. & Roepstorff, P. (1974) *Proc. Natl. Acad. Sci. U.S.A.* **71**, 2730–2733
Tiselius, A., Hjérten, S. O. & Levin, Ö. (1956) *Arch. Biochem. Biophys.* **65**, 132–155
Tishkoff, G. H., Williams, L. C. & Brown, D. M. (1968) *J. Biol. Chem.* **243**, 4151–4167

Physical Studies on Calcium Phosphate and Barium Citrate Used as Clotting-Factor Adsorbents

SUSAN L. M. DEACON and ROY M. HOWELL

Department of Biochemistry, Queen Elizabeth College, London W8 7AH, U.K.

Inorganic precipitates, containing Ca^{2+} or Ba^{2+} cations, have been used extensively in purification procedures for removing the vitamin K-dependent clotting factors from serum or plasma. The prothrombin group, which consists of factors II, VII, IX and X, contains the modified amino acid γ-carboxyglutamic acid which is responsible for the high cation-binding capacity of these proteins (Stenflo *et al.*, 1974).

We have examined the crystalline structure and charge of these inorganic compounds more closely, by scanning electron microscopy and particle electrophoresis, in an attempt to relate physical properties and chemical composition to their properties as clotting-factor adsorbents. The present investigation on the physical properties of calcium phosphate and barium citrate is similar to our previous study of $BaSO_4$ as a clotting-factor adsorbent (Howell & Deacon, 1975). The electrophoretic mobility of particles of various adsorbents was measured at 25°C in 0.02M-barbitone buffer over the pH range 2–11 by the procedure of Bangham *et al.* (1958).

Despite differences in morphology, various grades of $BaSO_4$, in the former study, gave similar pH–mobility curves, since the internal composition of the crystal remains relatively constant. In contrast, the present study showed a considerable shift in the shape

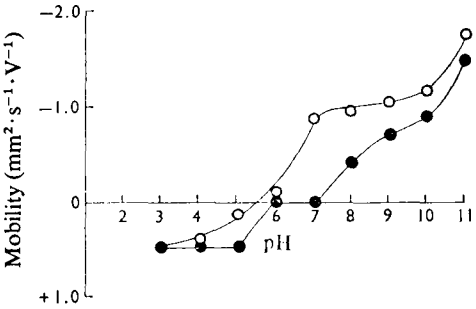

Fig. 1. *Electrophoretic mobility–pH curves for particles of calcium phosphate gel* (●) *and hydroxyapatite* (○)

Calcium phosphate gel was obtained from BDH, and hydroxyapatite was prepared by the method of Tiselius *et al.* (1956). Electrophoretic mobility was measured in 0.02M-barbitone buffer at 25°C. The sign indicates the polarity of particles and each point represents the average of twenty determinations.

of the pH–mobility curves from one adsorbent to another. This suggests that a change in the anionic component exerts more influence on particle charge than either of the cation components.

The origin of surface charge is complex and still rather obscure, but the sparingly soluble salts of calcium and barium can be separated into two groups according to the presence, or absence, of a charge reversal point in the pH–mobility curve. The charge characteristics for calcium phosphate gel (obtained from British Drug Houses, Poole, Dorset, U.K.) and hydroxyapatite (Tiselius *et al.*, 1956) (Fig. 1) are similar to those for various grades of $BaSO_4$ (Howell & Deacon, 1975). Apart from minor variations, the bimodal curve is composed of regions of positive and negative mobility on either side of a charge-reversal region. Factors contributing to this characteristic shape are adsorption of protons and free cations from the medium on the crystal lattice creating a positive potential and, as the pH increases, complexing of organic ions from the buffer causing a negative charge. The chemical composition of calcium phosphate is variable and in addition to the tricalcium orthophosphate form $Ca_3(PO_4)_2$, substitution of water and hydroxyl ions in the crystal lattice may result in the formation of brushite, $CaHPO_4$, $2H_2O$ or hydroxyapatite, $Ca_5(PO_4)_3OH$. Similarities between the curves in Fig. 1 suggest that the 'gel' form contains a considerable amount of hydroxyapatite, although vast differences in morphology were seen by scanning electron microscopy (Deacon & Howell, 1976). Micrographs of hydroxyapatite showed small well-defined crystal agglomerations in contrast with the irregular and totally amorphous particles seen for samples of calcium phosphate gel, specified for enzyme adsorption. The gel was freeze-dried to retain as much as possible of the hydrated structure for scanning electron microscopy, but the porous nature necessary for the separation of proteins was indiscernible, even at high magnifications. Both adsorbents take up serum proteins in relatively large amounts (Howell & Deacon, 1976), but, in contrast with high-density $BaSO_4$, neither has been found to induce factor X activation (Howell & Dupe, 1972).

The charge characteristics of a sparingly soluble salt can be altered when an ion is preferentially adsorbed on the surface (Reyerson *et al.*, 1947; Morimoto, 1964). In this respect some ions have a more pronounced effect than others and these are known as potential-determining ions. Morimoto (1964) and Reyerson *et al.* (1947) found that citrate ions altered the surface potential of $BaSO_4$ to a high negative value, which they attributed to the formation of a barium citrate complex on the surface. It has also been

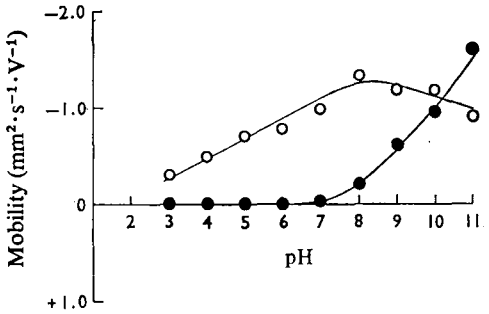

Fig. 2. *Electrophoretic mobility–pH curves for particles of calcium orthophosphate* (●) *and barium citrate* (○)

Calcium orthophosphate was obtained from BDH, and barium citrate (I.C.N.; K & K. Laboratories) was supplied commercially by Eastman-Kodak, Kirkby, Liverpool, U.K. Electrophoretic mobility details were as in Fig. 1.

---

reported (Morimoto, 1964; Tokiwa & Imamura, 1969) that the negative mobility of colloidal particles was unexpectedly high when phosphate ions were adsorbed from solution. These results reflect our observations on particles composed entirely of barium citrate or calcium orthophosphate which possess either a zero or negative potential over the pH range 3–11 (Fig. 2). These monophasic curves are typical of a surface in which the potential-determing group is either a carboxyl or phosphate ion (Goddard, 1970), and the indications are that the negatively charged tervalent citrate or phosphate ions suppress the positive potential of the bivalent cation in a crystal lattice. This suppression may prevent cation chelation by γ-carboxyglutamic acid residues, a theory supported by the finding that the vitamin K-dependent clotting factors show very little affinity for calcium orthophosphate or commercial barium citrate (Howell & Deacon, 1976).

Scanning-electron-microscopic studies on calcium orthophosphate showed irregular flaky amorphous plaques with crystalline edges visible at higher magnifications, and little or no activation has been reported for this adsorbent. The finding that commercial barium citrate is composed of large non-porous crystals, reminiscent of high-density $BaSO_4$, was a surprising anomaly, since it has not been reported to induce activation of factor X. A possible explanation may be that since barium citrate is precipitated *in situ* during the purification procedures (Radcliffe & Nemerson, 1975), incorporation of clotting factors into the crystal lattice (Tishkoff *et al.*, 1968) protects the protein from activation.

We are indebted to the Stereoscan Microscope Unit at Bedford College, University of London, for the photomicrographs and to Professor A. M. James of the Chemistry Department at the above College for much helpful advice. This investigation was supported by a Science Research Council studentship to S. L. M. D.

Bangham, A. D., Flemans, R., Heard, D. H. & Seaman, G. V. F. (1958) *Nature (London)* **182**, 642–644
Deacon, S. L. M. & Howell, R. M. (1976) *Thromb. Haemostas.* in the press
Goddard, G. H. (1970) Ph.D. Thesis, University of London
Howell, R. M. & Deacon, S. L. M. (1975) *Thromb. Diath. Haemorrh.* **33**, 256–270
Howell, R. M. & Deacon, S. L. M. (1976) *Biochem. Soc. Trans.* **4**, 718–720
Howell, R. M. & Dupe, R. J. (1972) *Thromb. Diath. Haemorrh.* **28**, 306–316
Morimoto, T. (1964) *Bull. Chem. Soc. Jpn.* **37**, 386–392
Radcliffe, R. & Nemerson, Y. (1975) *J. Biol. Chem.* **250**, 388–395
Reyerson, L. H., Kolthoff, I. M. & Coad, K. (1947) *J. Phys. Colloid Chem.* **51**, 321–332

Stenflo, J., Fernlund, P., Egan, W. & Roepstorff, P. (1974) *Proc. Natl. Acad. Sci. U.S.A.* **71**, 2730–2733
Tiselius, A., Hjérten, S. & Levin, Ö. (1956) *Arch. Biochem. Biophys.* **65**, 132–155
Tishkoff, G. H., Williams, L. C. & Brown, D. M. (1968) *J. Biol. Chem.* **243**, 4151–4167
Tokiwa, F. & Imamura, T. (1969) *J. Am. Oil Chem. Soc.* **46**, 280–284

Apparent Alterations of Erythrocyte Acetylcholinesterase and other Membrane Proteins in Duchenne Muscular Dystrophy: a further example of a Generalized Membrane Defect Associated with Hereditary Muscular Dystrophy

P. K. DAS,* D. GRAESSLIN† and H. W. GOEDDE*

**Institute of Human Genetics and †Department of Clinical and Experimental Endocrinology, University of Hamburg, Hamburg, Federal Republic of Germany*

The primary cause of genetically determined muscular dystrophy remains obscure. Evidence in favour of purely myopathic (Walton, 1969), neoronal (Gallup & Dubowitz, 1973), vascular (Hathway *et al.*, 1970) and autoimmune (Jasmin & Bokdawala, 1970) mechanisms has been put forward. Attempts to find a mutant protein have not yet produced any substantial results. The increase that occurs in certain intracellular enzymes, such as creatine kinase, in the serum of patients with Duchenne muscular dystrophy (Pennington, 1969) may only indicate that the organization of plasma membrane with respect to its permeability is altered. A postulation, based on various observations (Das *et al.*, 1971*a*,*b*; Rodan *et al.*, 1974; Roses *et al.*, 1975), strongly points to a generalized membrane defect as the fundamental problem in this disorder.

To assimilate the different concepts on the lesion underlying this disease, we chose to investigate a functional protein that may be involved in maintaining normal neuromuscular activity. Acetylcholinesterase plays an important role in neuromuscular transmission (Nachmansohn, 1959). Since erythrocyte acetylcholinesterase appears to be similar (at least with respect to its substrate specificity) to the acetylcholinesterase at the neuromuscular junction and because the material is easily available, we have previously compared some properties of the enzyme in normal and diseased mice (Das *et al.*, 1971*a*,*b*). In this presentation, we report further evidence confirming earlier observations on the altered properties of Triton X-100-solubilized erythrocyte acetylcholinesterase, which indicates membrane abnormality in Duchenne muscular dystrophy. Other erythrocyte-membrane enzymes (Brown *et al.*, 1967; Roses *et al.*, 1975) have also been reported to be altered in this disease. The possibility remains that the membrane defect in Duchenne muscular dystrophy can also be demonstrated when membrane constituents other than those already reported are investigated. For example, in spite of the reported similar protein pattern on sodium dodecyl sulphate/polyacrylamide-gel disc electrophoresis (Roses *et al.*, 1975), we have observed different protein patterns of Triton-solubilized erythrocyte 'ghosts' from patients with Duchenne muscular dystrophy, on slab gel electrofocusing.

P. K. D. is indebted to the Muscular Dystrophy Group, U.K., the Hong Kong University Research Grant no. 158/229 for financial support and to Alexander von Humboldt-Stiftung for the award of Senior Fellowship, to continue this work. We thank Professor W. J. Bradeley, Professor Bethlem, Dr. J. O'Brian, Professor R. Beckmann, Professor D. Seitz, for supplying the samples.

Brown, H. D., Chattapodhya, S. K. & Patel, A. B. (1967) *Science* **57**, 1577–1578
Das, P. K., Watts, R. L. & Watts, D. C. (1971*a*) *Biochem. J.* **123**, 24P–25P
Das, P. K., Watts, D. C. & Coles, H. M. T. (1971*b*) *Abstr. FEBS Meet. 7th*, 307
Gallup, B. & Dubowitz, V. (1973) *Nature (London)* **243**, 287–289
Hathway, P. W., Engel, W. K. & Zwellger, H. (1970) *Arch. Neurol.* **22**, 365–378
Jasmin, G. & Bokdawala, F. (1970) *Rev. Can. Biol.* **29**, 197–201